

Solution Requirements

Date	5 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users (Farmers/Admin) can securely register, log in, and log out using a username and password	High
2	Authorization levels	Farmers can access crop prediction and reports. Admin can manage users, datasets, and system information.	High
3	External interfaces	The system integrates with the Machine Learning model, crop dataset, and weather APIs (if available)	Medium

		The system receives user input (soil nutrients, pH, temperature, humidity, rainfall), processes it through the ML model, and returns the recommended crop.	
4	Transactions processing	The system accepts soil nutrients (N, P, K), pH, temperature, humidity, and rainfall, processes the data through the ML model, and displays the recommended crop.	Medium
5	Reporting	The system generates crop prediction results, user history, and reports that can be viewed or downloaded by the user.	Medium
6	Business rules, etc.	The system validates all required input fields before prediction. Recommendations are generated only from the trained machine learning model.	High
7	Compliance to laws or regulations	User information is stored securely, and the system follows basic data privacy and security practices	Medium
8	<i>Other</i>	Users can view previous predictions and receive crop recommendations with basic farming guidance	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Crop prediction should be generated quickly after the user submits the input values.	Prediction displayed within 2 seconds
2	Scalability	The system should support an increasing number of users and prediction requests without performance degradation.	Supports 1,000+ users
3	Security & Data Privacy	User credentials and personal information should be protected from unauthorized access.	Passwords stored securely and HTTPS used where applicable
4	Reliability & Availability	The application should provide accurate predictions and remain available during normal usage.	99% system availability
5	Usability & Accessibility	The interface should be simple, responsive, and easy to use on desktop and mobile devices.	Responsive design with user-friendly navigation
6	<i>Other</i>	The system should be maintainable and portable so that it can be updated and deployed on different operating systems.]	Modular code compatible with Windows and Linux